Brent Crude Volatility and Value-at-Risk Report

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Executive Summary

This ‘Brent crude volatility and Value-at-Risk’ report uses daily price data from Yahoo Finance to analyse how volatile Brent crude oil futures are and walks through the building and testing of models that can be used to forecast that volatility. The forecasts are then used to calculate a Value-at-Risk (VaR) figure, which estimates the amount that could be lost in one day. The data exploration, modelling and testing took place in Google Colab, with the GARCH model built using the arch package and the random forest built using packages from SciKit Learn.

The results show that Brent volatility is not constant, with calm and turbulent periods that cluster together. Daily returns also have very fat tails, with an excess kurtosis of ~14.55. Out of the models tested on 1,195 unseen trading days from 2022 onwards, the GARCH model was the most accurate, beating the best simple baseline by ~5.7% on RMSE and ~2.3% on MAE. The random forest did not beat GARCH. The GARCH model also produced the most reliable VaR, with 61 breaches of the 95% VaR limit compared with ~60 expected. The baseline and random forest were both breached 73 times. It is worth noting that volatility models measure the size of price moves and not their direction, so they cannot be used on their own as a trading signal.

Hypothesis

When forecasting the volatility of Brent crude, I believe that the GARCH model will perform better than both the simple baselines and the random forest. This hypothesis is based on volatility clustering, where large price moves tend to follow large price moves, and on the fact that GARCH is a model that was designed to capture this behaviour directly. I also believe that the random forest will struggle during the sharpest spikes in volatility, as a decision tree can only predict averages of values that it has already seen in the training data.

Similarly, I believe that a more accurate volatility forecast should lead to a more reliable Value-at-Risk estimate, with a 95% VaR limit being breached on close to 5% of days. If the loss limits are breached much more often than this, the model is understating the risk, which is a problem for any risk team relying on the number.

Introduction

Brent crude oil is one of the most widely traded commodities in the world, and it is traded through futures contracts, which are agreements to buy or sell oil at a fixed price on a future date. The price of oil does not move by the same amount every day. Some periods are calm and others are turbulent, and the size of the daily swings (the volatility) changes over time. Traders and risk managers need to know how large those swings are likely to be, because it determines how much money could be lost on a bad day.

This report asks two questions. The first is whether the volatility of Brent crude over the next five trading days can be forecast, and which method does it best. The second is whether a better volatility forecast produces a more reliable estimate of the amount that could be lost in a day. To answer these, I trained a GARCH model and a random forest model using the first twelve years of data (2010 to 2021) and then tested both of them, alongside two simple baselines, on data from 2022 onwards that was hidden before the models were fitted.

The dataset was downloaded from Yahoo Finance using the yfinance library, under the ticker BZ=F, which is a continuous series of the front-month Brent futures contract. It contains 4,187 daily closing prices from 4 January 2010 to 9 October 2026. The closing price was used to calculate the daily log return, and from this the following fields were built for the models: the 5-day, 21-day and 63-day rolling volatility, the absolute return for the day, the 5-day return and the target (the volatility over the next 5 trading days).

While the results show that the GARCH model is a good fit for Brent over this period, I am not suggesting that these results apply to other commodities, or that this model can predict the direction of the oil price. Volatility models only use past prices, so they cannot anticipate news such as supply disruptions or geopolitical events, and a sudden shock can push actual volatility well above the forecast. To use the model as part of a real risk process, it would need to be tested on rollover-adjusted price data, and compared with further volatility models.

Methodology

To access the data, the yfinance library was used within Google Colab to download the daily prices for BZ=F from Yahoo Finance. The end date was fixed at 10 October 2026 (the end date in yfinance is exclusive), so that an incomplete final trading day was not included and the results can be reproduced. The closing price was used to calculate the log return for each day, which is the natural log of the price today divided by the price yesterday. Log returns were used because they can be added together across multiple days to give the overall move.

The features were then built from the returns. These were the rolling standard deviation of the returns over 5, 21 and 63 trading days (the volatility), the absolute return for the day, and the return over the last 5 days. The target was the standard deviation of the next 5 daily returns. Rows with missing values were removed, which left 4,119 rows.

The data was split by date and not randomly, because a real forecaster only ever knows the past. Data up to 31 December 2021 was used for training (2,924 rows) and data from 2022 onwards was hidden from the models and used for testing (1,195 rows). Two simple baselines were tested first, where the volatility over the next 5 days is assumed to equal either the last 5 days or the last 21 days. These involve no fitting, so any model built after this must beat them to be worth the extra complexity. The forecasts were scored using the root mean squared error (RMSE) and the mean absolute error (MAE).

The first model was a GARCH(1,1) model with a Student-t error distribution, fitted using the arch package on the training data only. The returns were multiplied by 100 to help the model fit reliably, and a Student-t distribution was used because the exploration showed that the returns have fat tails. To forecast the test period without the model learning from it, the fitted parameters were fixed and the model only updated its variance estimate each day using that day's actual return. The variance forecasts for the next 5 days were averaged before taking the square root, because variances add across days but standard deviations do not.

The second model was a random forest, imported from SciKit Learn, which was used to test for non-linear relationships that GARCH cannot capture. It was trained on the five features with 300 trees, a maximum depth of 5 and a minimum of 20 days in each final group, which limits how complex each tree can get. The feature importances were then used to see which inputs the forest relied on, and all trading days were split into five equal groups by the past week's return to check whether volatility differs after price falls and price rises.

For the Value-at-Risk (VaR) backtest, each model's volatility forecast was turned into a 95% loss limit for a position of £100,000, using a multiplier of -1.645 from the normal distribution. This limit was compared with the actual return on the following day, and if the loss was worse than the limit it was counted as a breach. The breach counts were tested using the Kupiec test, which checks whether the breach rate is consistent with the 5% target. For the GARCH model, the chance of a breach on the day after a breach was also compared with the chance of a breach after a day with no breach, to check whether the breaches bunch together.

Finally, the GARCH model was refitted on the full history and used to forecast the next 21 trading days (one month).

Definitions

Important terms that should be explained can be found below:

Futures contract: An agreement to buy or sell an asset, in this case Brent crude oil, at a fixed price on a future date. The series used in this report is a continuous front-month series, which means the price jumps when it moves from one contract to the next.

Log return: The natural log of the price today divided by the price yesterday. Log returns can be added together across several days to give the overall move.

Volatility: The standard deviation of the daily log returns over a period, which measures how large the daily price swings are. Multiplying the daily figure by the square root of 252 (the approximate number of trading days in a year) converts it to an annualised figure.

Volatility clustering: The tendency for large price moves to be followed by large price moves, and small moves by small moves, so calm and turbulent periods come in groups.

Excess kurtosis: A measure of how heavy the tails of a distribution are compared with a normal distribution, where 0 means normal tails. A higher value means extreme moves happen more often than a normal model predicts.

RMSE and MAE: Two measures of forecast error where lower is better. The mean absolute error (MAE) is the average size of the miss, while the root mean squared error (RMSE) squares the misses before averaging, so it punishes large misses more heavily.

GARCH(1,1): A statistical model for volatility where tomorrow's variance is a weighted mix of a long-run baseline level, yesterday's squared return and yesterday's variance estimate. It captures both volatility clustering and the tendency for volatility to drift back to a long-run average.

Random forest: A machine learning algorithm that uses many decision trees to make better predictions. Each tree looks at different random parts of the data and their results are averaged, which makes it an ensemble learning technique. This helps in improving accuracy and reducing errors.

Value-at-Risk (VaR): A loss limit that should only be exceeded on a given share of days. A 95% VaR should be breached on about 1 day in 20.

Breach: A day where the actual loss is worse than the VaR limit.

Kupiec test: A hypothesis test that checks whether the number of breaches is consistent with the target rate. The null hypothesis is that the true breach probability is 5%, and a p-value below 0.05 rejects the model.

Results

The dataset contained 4,187 daily closing prices for Brent crude from 4 January 2010 to 9 October 2026. Volatility is clearly not constant over this period. It was low through 2012 to 2014, rose sharply from late 2014, peaked at more than 150% annualised in spring 2020 and spiked again in 2022 and 2026. The daily log returns are also not normally distributed, with a skewness of ~-0.95 and an excess kurtosis of ~14.55. This means that large falls are bigger and more common than large rises, and that extreme moves happen far more often than a normal model allows. The five largest daily moves were between ~14% and ~28% in log terms, with four of them in March and April 2020 and one on 1 April 2026 (see Figures 1 and 2).

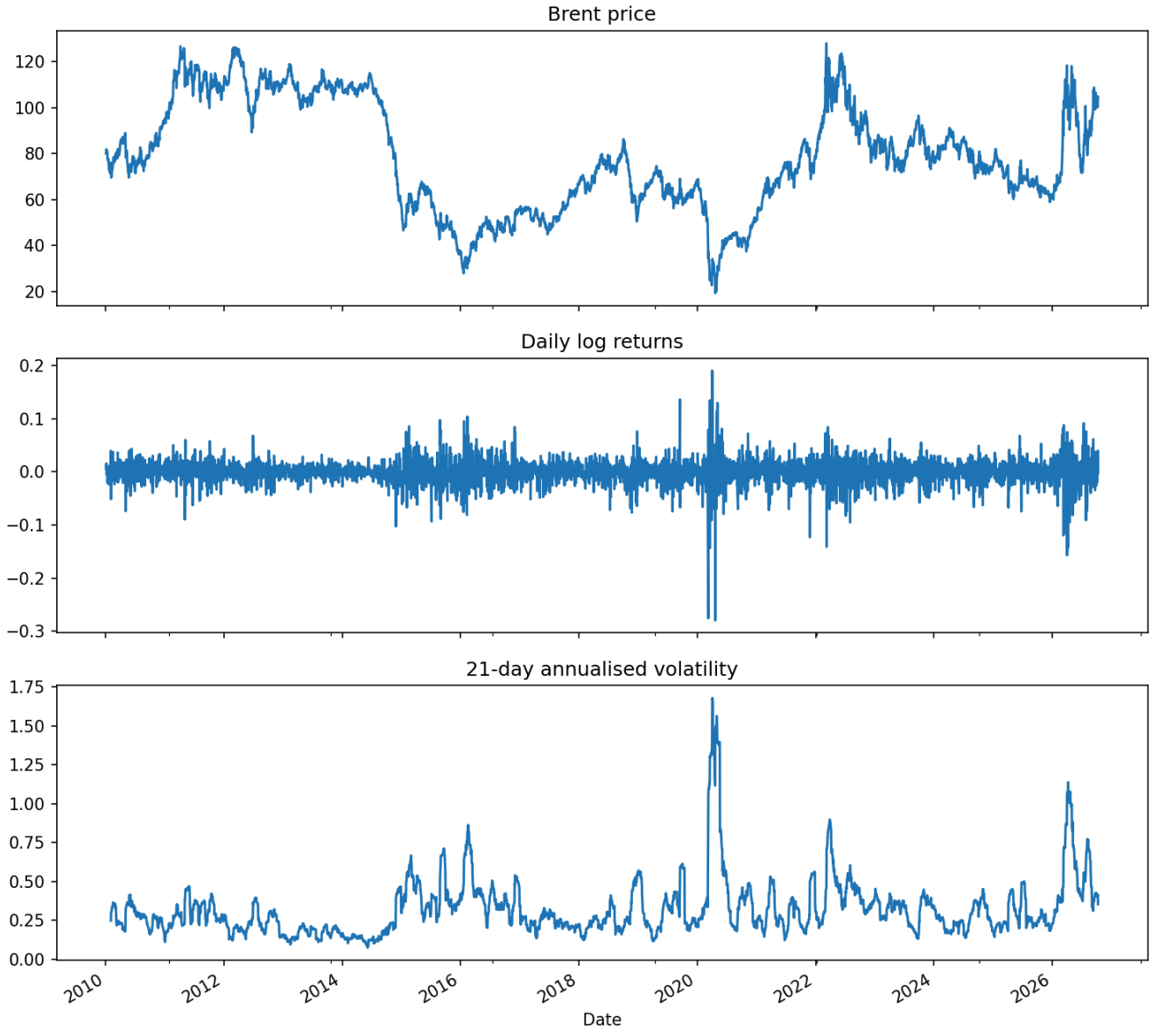


Figure 1: Brent price, daily log returns and 21-day annualised volatility, 2010 to 2026.

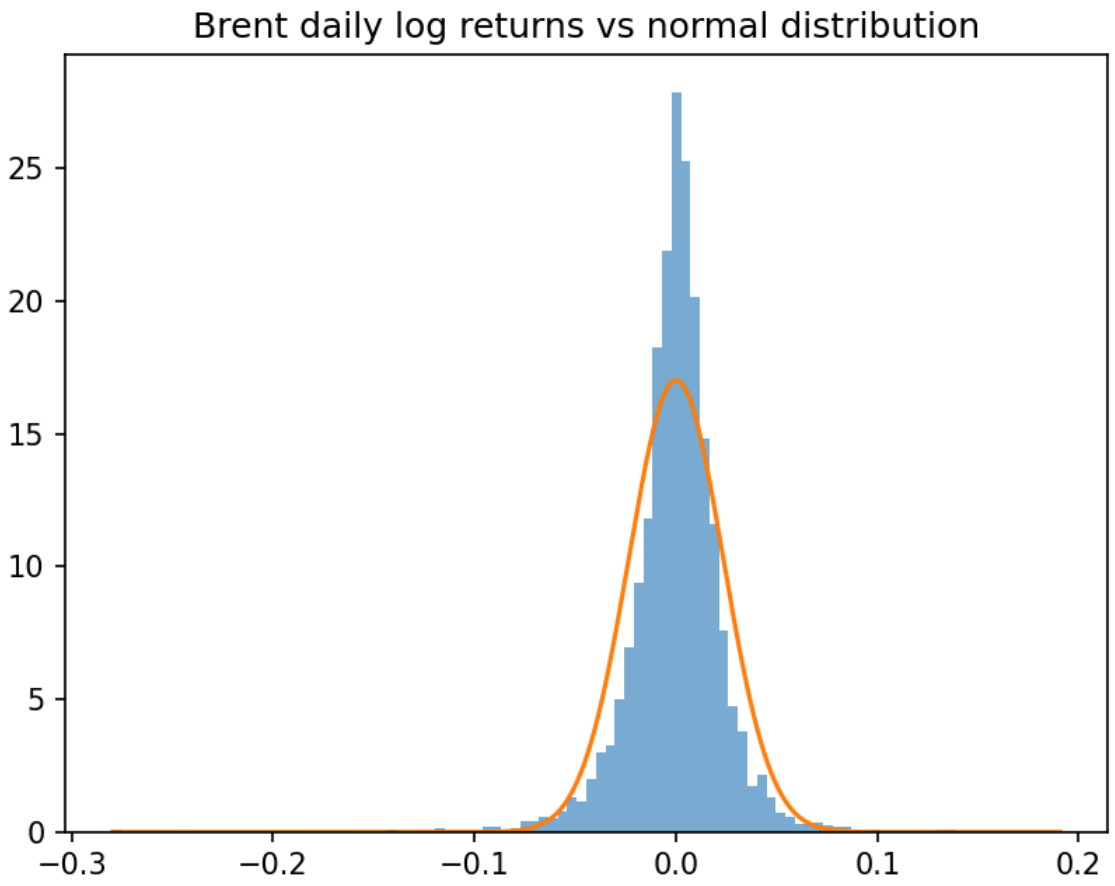


Figure 2: Brent daily log returns against a normal distribution with the same mean and standard deviation.

For the baselines, the 5-day rule had an RMSE of 0.01343 and an MAE of 0.00932, while the 21-day rule had an RMSE of 0.01160 and an MAE of 0.00818. The 21-day rule was more accurate on both measures, so these scores became the benchmark that the models had to beat.

The first model was the GARCH(1,1) model with a Student-t distribution, fitted on 2,986 training observations. The sum of alpha (0.088) and beta (0.904) was ~0.992, which is very close to 1, meaning volatility shocks fade slowly with a half-life of ~85 trading days. The parameters imply a long-run volatility of ~2.8% a day, or ~44% annualised. The nu value of ~4.5 confirms the fat tails, and the volatility parameters were all statistically significant at the 1% level. On the 1,195 unseen test days, GARCH scored an RMSE of 0.01094 and an MAE of 0.00799, which beats the best baseline by ~5.7% on RMSE and ~2.3% on MAE. This is a modest improvement, which is realistic because recent volatility is already a strong predictor of near-future volatility. Looking at the forecast chart, GARCH tracks the underlying level of volatility well and reacts to shocks slightly faster than the 21-day rolling window, but it underestimates the sharpest peaks in 2022 and 2026 (see Figure 3).

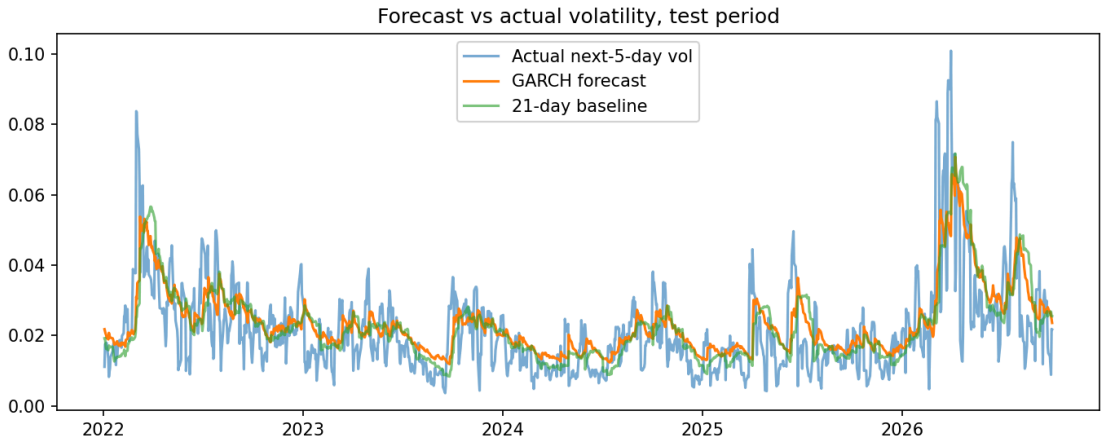


Figure 3: Actual next-5-day volatility, the GARCH forecast and the 21-day baseline over the test period (2022 onwards).

The second model was the random forest, which scored an RMSE of 0.01190 and an MAE of 0.00812. This beats the 5-day baseline but not GARCH, and it is roughly level with the 21-day baseline, with a marginally better MAE and a slightly worse RMSE. This suggests that the forest makes larger misses during volatility spikes, which is likely because trees cannot predict beyond the range of values they saw in training. Looking at the feature importances, the 21-day volatility was the most important feature by a large margin with a score of 0.687, which is the same quantity used in the best baseline, so the forest had little new information to add. The next features were the 5-day return (0.114), the 5-day volatility (0.108) and the 63-day volatility (0.084), while the absolute return for the day scored only 0.006.

The 5-day return ranking second is notable because it measures direction and not the size of the swings. To check this, all trading days were split into five equal groups by the past week's return. The average volatility over the following 5 days was lowest after a flat week (0.0153) and higher after large moves in either direction, forming a U-shape. The pattern is also asymmetric, with volatility after a big fall (0.0248) being ~14% higher than after a big rise (0.0218), and volatility after an ordinary fall (0.0175) being ~10% higher than after an ordinary rise (0.0159). This is known as the leverage effect, and a standard GARCH model treats up and down moves the same, so it cannot capture it.

For the VaR backtest, a 95% loss limit should be breached on about 60 of the 1,195 test days. The GARCH model was breached on 61 days (a breach rate of ~5.1%), while the baseline and the random forest were both breached on 73 days (~6.1%), meaning their loss limits were about 22% too tight. The Kupiec test did not reject any of the models at the 5% level. The GARCH model had a likelihood ratio of 0.03 and a p-value of 0.869, so its breach count is almost exactly what chance alone would produce. The baseline and random forest each had a likelihood ratio of 2.90 and a p-value of 0.089, which is just above

the 0.05 cut-off, so there is some evidence that they understate risk but not enough to reject them. Not rejecting a model does not prove it is correct, since with only about 60 expected breaches the test has limited power to detect modest errors.

For the GARCH breaches, the chance of a breach on the day after a breach was ~6.6%, compared with ~5.0% after a day with no breach. This is a slight hint of clustering, but the first figure is only 4 breaches out of 61 days that followed a breach, so one or two breaches either way would move it noticeably. This is a weak indication and not evidence that the breaches bunch together.

Finally, the GARCH model was refitted on the full history and used to forecast the next 21 trading days. The forecast is an average daily volatility of ~2.31%, which is ~36.7% annualised and below the long-run level of ~44%, meaning the model sees current conditions as calmer than average. The standard deviation of the one-month return is ~10.6%, so roughly two-thirds of the time the price move over the month should fall within ±10.6% of the starting price. This describes the width of the range of likely outcomes and does not say which way the price will move. It is also worth noting that this 21-day forecast has not been backtested, since the backtest only covers 5-day forecasts.

Conclusion

This report shows that the volatility of Brent crude changes over time and can be forecast with reasonable success from recent price history. The GARCH(1,1) model with a Student-t distribution was the most accurate of the models tested, and it also produced the most reliable Value-at-Risk estimate, with a breach count that was almost exactly the number expected from a 95% loss limit. This supports the hypothesis that GARCH would perform best, and that a better volatility forecast leads to a more reliable VaR.

The random forest did not beat the simpler statistical model. Its most important input was the same 21-day volatility that the baseline already used, and trees cannot predict beyond the range of values they saw in training. The improvement from GARCH was also modest, and the ranking comes from a single train and test split, so the results should not be taken as proof that GARCH will be the best model for other commodities.

There are several limitations that are worth further investigation. The BZ=F series is a continuous front-month contract, so some of the recorded returns are rollover jumps and not real price moves, which can inflate the volatility estimates. The VaR applies a 5-day volatility forecast to a 1-day loss and assumes normally distributed returns, even though the returns have fat tails. The 21-day forecast has not been backtested, and GARCH underestimated the sharpest spikes because it can only respond once large moves have appeared in the data. Finally, a volatility forecast describes the size of likely price moves and not their direction, so it should be used to manage risk and not as a trading signal.

To further this project, the next step is to add further models and compare them fairly with GARCH, including EWMA, GJR-GARCH (to capture the leverage effect) and a HAR-style regression. The VaR could also be improved by using the Student-t distribution and by testing a 99% confidence level, and the analysis should be repeated using rollover-adjusted prices and on other assets such as WTI crude, natural gas and equity indices.

References

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